

Introduction to Contract Theory and Auctions

Mathematical Tools

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

ECON 403/503
Fall 2021

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathcal{D} \subset \mathbb{R}^n$. A constrained optimization problem is

$$\max f(x) \quad \text{subject to } x \in \mathcal{D}.$$

- f is the **objective function**.
- $\mathcal{D}$ is the **constraint set**.
- A solution to this problem is $x \in \mathcal{D}$ such that

$$f(x) \geq f(y) \quad \text{for all } y \in \mathcal{D}.$$

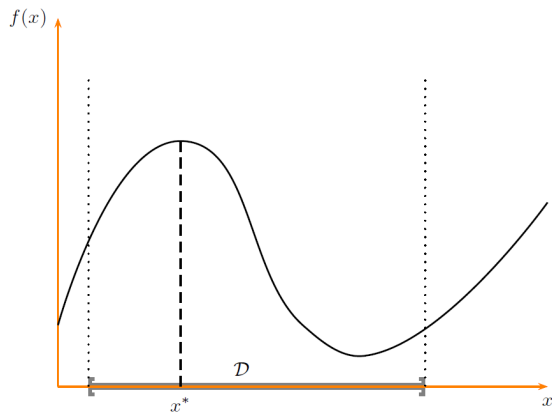
Such an x is called a **maximizer**.

- The set of maximizers is denoted

$$\arg \max \{f(x) | x \in \mathcal{D}\}$$

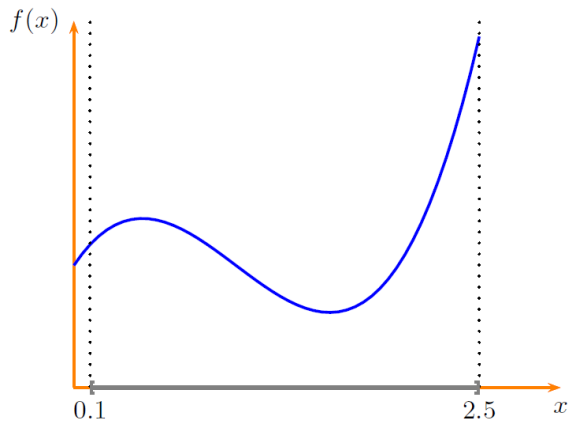
- The notation is similar for minimization problems.

A Graphical Example



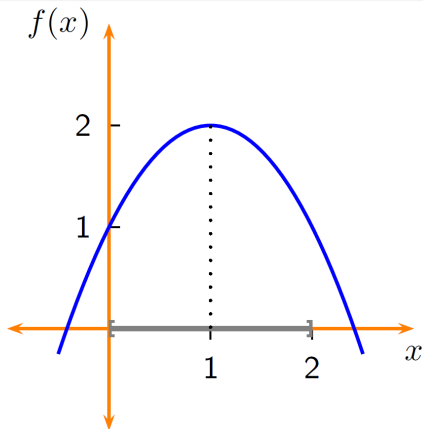
Example A.1

$\max x^3 - 3x^2 + 2x + 1$ subject to $0.1 \leq x \leq 2.5$.



Example A.2

$\max -(x - 1)^2 + 2$ s.t. $x \in [0, 2]$.



Theorem A.3 (Weierstrass Theorem)

Let $\mathcal{D} \subset \mathbb{R}^n$ be compact and $f : \mathcal{D} \rightarrow \mathbb{R}$ be a continuous function on $\mathcal{D}$. Then f attains a maximum and a minimum on $\mathcal{D}$.

Theorem A.4

A set $\mathcal{D}$ is **compact** if and only if it is closed and bounded.

Definition A.5

A set $\mathcal{D}$ is **bounded** if there exists $k > 0$ such that $\|x\| < k$ for each $x \in \mathcal{D}$.

Here $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$ is the **Euclidean norm**. If x is a real number, then $\|x\|$ is simply the absolute value $|x|$.

Example A.6

$\mathcal{D} = [0, 2]$. For any $x \in [0, 2]$, $\|x\| = \sqrt{x^2} = |x|$. Therefore, we can take $k = 3$, and hence this set is bounded.

Example A.7

$\mathcal{D} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$. Take any $(x, y) \in \mathcal{D}$ and note that $\|(x, y)\|^2 = x^2 + y^2 \leq 1$. Therefore, $\|(x, y)\| \leq 1$. Thus, this set is bounded.

Definition A.8

A set $\mathcal{D} \subset \mathbb{R}^n$ is **closed** if and only if for all sequences (x_k) such that $x_k \in \mathcal{D}$ for each k and $\lim_{k \rightarrow \infty} x_k = x$, we have that $x \in \mathcal{D}$.

The **Euclidean distance** between $x, y \in \mathbb{R}^n$ is given by

$$d(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2}$$

Definition A.9

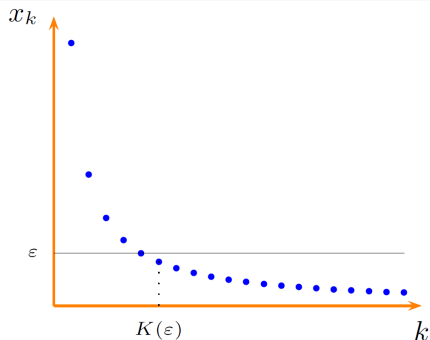
A sequence (x_k) in $\mathbb{R}^n$ is said to converge to a limit x (written as $\lim_{k \rightarrow \infty} x_k = x$ or $x_k \rightarrow x$) if for all $\epsilon > 0$, there exists an integer $K(\epsilon)$ such that for all $k \geq K(\epsilon)$, we have $d(x_k, x) < \epsilon$.

Definition A.10

A set $\mathcal{D} \subset \mathbb{R}^n$ is **open** if, for any point $x \in \mathcal{D}$, there exists $\epsilon > 0$ such that, given any point $y \in \mathbb{R}^n$ satisfying $d(x, y) < \epsilon$, then y also belongs to $\mathcal{D}$.

Example A.11

Consider the sequence (x_k) , where $x_k = \frac{1}{k}$ for $k \in \mathbb{N}$. This sequence converges to 0. To see this, take any $\epsilon > 0$ and let $K(\epsilon)$ be the smallest integer larger than $\frac{1}{\epsilon}$. Then, $k \geq K(\epsilon)$ implies that $d(x_k, 0) = 1/k \leq 1/K(\epsilon) < \epsilon$.



Example A.12

Claim: $\mathcal{D} = [0, 1]$ is closed. Proof: Take any sequence $(x_k) \rightarrow x$, such that $x_k \in [0, 1]$ for all k . This implies that for any $\epsilon > 0$, there exists an integer $K(\epsilon)$ such that for all $k \geq K(\epsilon)$, we have that $d(x_k, x) = |x_k - x| < \epsilon$, or $x_k - \epsilon < x < x_k + \epsilon$. Since $x_k \geq 0$ and $x_k \leq 1$, this implies that for any $\epsilon > 0$, $-\epsilon < x < 1 + \epsilon$. Since ϵ is arbitrary this implies that $0 \leq x \leq 1$, which implies that $x \in \mathcal{D}$.

Example A.13

Claim: $\mathcal{D} = [0, 1)$ is not closed. Proof: By counterexample, take the sequence (x_k) , where $x_k = 1 - 1/k$, for $k \in \mathbb{N}$, and note that $x_k \in \mathcal{D}$ for all k . Also $(x_k) \rightarrow 1$ but $1 \notin \mathcal{D}$.

Definition A.14

A function $f : S \rightarrow T$ where $S \subset \mathbb{R}^n$ and $T \subset \mathbb{R}^\ell$ is **continuous** at $x \in S$ if for all $\epsilon > 0$, there is a $\delta > 0$ such that $y \in S$ and $d(x, y) < \delta$ implies $d(f(x), f(y)) < \epsilon$. Equivalently, f is continuous at $x \in S$ if for all (x_k) such that $x_k \in S$ for each k and $x_k \rightarrow x$, we have $f(x_k) \rightarrow f(x)$. f is continuous on S if it is continuous at each $x \in S$.

Definition A.15

$f : S \rightarrow \mathbb{R}$, where S is an open subset of $\mathbb{R}$, is differentiable at $x \in S$ if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists, in which case this limit is the derivative of f at x , denoted $f'(x)$ or $Df(x)$.

Definition A.16

Let $f : S \rightarrow \mathbb{R}$, where S is an open subset of $\mathbb{R}^n$. The j th partial derivative of f exists at $x \in S$ and is equal to

$$\lim_{\epsilon \rightarrow 0} \frac{f(x_1, \dots, x_{j-1}, x_j + \epsilon, x_{j+1}, \dots, x_n) - f(x_1, \dots, x_{j-1}, x_j, x_{j+1}, \dots, x_n)}{\epsilon} = \frac{\partial f}{\partial x_j}(x)$$

if that limit exists. Partial derivatives are also denoted as: $\frac{\partial f}{\partial x_j}(x) = f_j(x)$.

If $f : S \rightarrow \mathbb{R}$, where S is an open subset of $\mathbb{R}^n$, and f is differentiable on S , then the derivative of f at x is written as

$$Df(x) = \left[\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_n} \right]$$

Note that Df itself defines a function from S to $\mathbb{R}^n$. $Df(x)$ is also called the **gradient of f at x** .

- If Df is differentiable, i.e., each partial derivative $f_i : S \rightarrow \mathbb{R}$ is differentiable at x , we denote the partial derivatives of f_i with respect to i and $j \neq i$ as

$$\frac{\partial^2 f(x)}{\partial x_i^2} \quad \text{and} \quad \frac{\partial^2 f(x)}{\partial x_j \partial x_i}$$

- We say that f is twice differentiable at x , with second derivative $D^2f(x)$ given by

$$D^2f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}$$

The matrix $D^2f(x)$ is called the **Hessian matrix at x** .

- If a function is (twice) continuously differentiable we say that it is $\mathcal{C}^1$ ($\mathcal{C}^2$). For a $\mathcal{C}^2$ function

$$\frac{\partial^2 f(x)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x)}{\partial x_j \partial x_i}.$$

Example A.17

Let $f(x) = -(x - 1)^2 + 2$. Then $Df(x) = f'(x) = -2(x - 1)$ and $D^2f(x) = f''(x) = -2$.

Example A.18

Let $f(x, y) = x^2 - y + xy$. Then, $Df(x, y) = [2x + y, -1 + x]$ and

$$D^2f(x) = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$$

- We can extend the idea of the derivative to functions $F : S \rightarrow T$, where $S \subset \mathbb{R}^n$ and $T \subset \mathbb{R}^\ell$.
- Such a function F can be thought as having ℓ -component functions $f^i : S \rightarrow \mathbb{R}$, $i = 1, \dots, \ell$, that is, for $x \in S$, $F(x) \mapsto (f^1(x), f^2(x), \dots, f^\ell(x)) \in T$.
- Partial derivatives, if they exist, are defined as in definition A.16: the partial derivative of the k component function with respect to its i th argument is denoted $\frac{\partial f^k(x)}{\partial x_i} = f_i^k(x)$.

Definition A.19

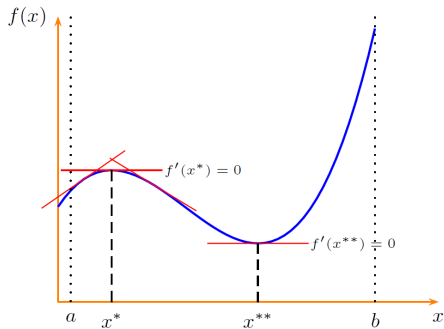
If $F : S \rightarrow T$, where S is an open subset of $\mathbb{R}^n$, and F is differentiable on S , then the derivative of F at x is given by the matrix:

$$DF(x) = \begin{bmatrix} \frac{\partial f^1(x)}{\partial x_1} & \cdots & \frac{\partial f^1(x)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f^\ell(x)}{\partial x_1} & \cdots & \frac{\partial f^\ell(x)}{\partial x_n} \end{bmatrix}$$

$DF(x)$ is also called the **Jacobian of F at x** .

A simple case

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and consider the problem $\max_{[a,b]} f(x)$



We call a point x^* such that $f'(x^*) = 0$ a critical point.

Theorem A.20

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and suppose $a < x^* < b$ is a local maximum (minimum) of f on $[a, b]$ and that f is differentiable at x^* . Then $f'(x^*) = 0$.

- The above are known as **first order conditions**.
- Only necessary for interior local optima.
 - Not necessary for global optima.
 - Not sufficient for local optima.
- To distinguish between interior local maximum and minimum you can use **second order conditions**.

Theorem A.21

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and suppose $a < x^* < b$ is a local maximum (minimum) of f on $[a, b]$ and that f is twice differentiable at x^* . Then $f''(x^*) \leq 0$ ($f''(x^*) \geq 0$).

Recipe for solving the simple case

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function and consider the problem $\max_{x \in [a, b]} f(x)$. If the problem has a solution, then it can be found by the following method:

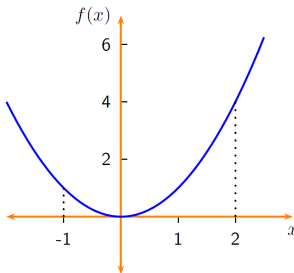
- ① Find all critical points: i.e., $x^* \in [a, b]$ such that $f'(x^*) = 0$.
 - ② Evaluate f at all critical points and at boundaries a and b .
 - ③ The one that gives the highest f is the solution.
- We can use Weierstrass theorem to determine if there is a solution.
 - Note that if $f'(a) > 0$ (or $f'(b) < 0$) , then the solution cannot be at a (or b).

Example A.22

$$\max x^2 \text{ s.t. } x \in [-1, 2].$$

Solution

x^2 is continuous and $[-1, 2]$ is closed and bounded, and hence compact. Therefore, by Weierstrass theorem the problem has a solution. $f'(x) = 2x$ is solved at $x = 0$, which is the only critical point. We have $f(0) = 0$, $f(-1) = 1$, and $f(2) = 4$. Therefore, 4 is the global maximum.



Example A.23

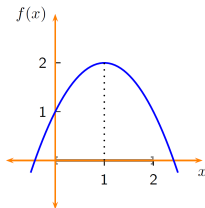
$$\max -(x - 1)^2 + 2 \text{ s.t. } x \in [0, 2].$$

Solution

f is continuous and $[0, 2]$ is compact. Therefore, the problem has a solution.

$f'(x) = -2(x - 1) = 0$ is solved at $x = 1$, which is the only critical point. We have

$f(1) = 2, f(0) = 1, f(2) = 1$. Therefore, 1 is the only global maximum. Note that $f'(0) > 0$ and $f'(2) < 0$ and hence we could have eliminated 0 and 2 as candidates.



What happens to the solution if the constraint set is $[-1, 0.5]$?

- Generalizes to $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and the problem is

$$\max f(x) \quad \text{subject to } x \in \mathcal{D}$$

- Find critical points $x^* \in \mathcal{D}$ such that $Df(x^*) = 0$.
 - Evaluate f at the critical points and the boundaries of $\mathcal{D}$.
 - Choose the one that give the highest f .
- Important to remember that solution must exist for this method to work.
- In more complicated problems evaluating f at the boundaries of $\mathcal{D}$ could be difficult.
- For such cases we have the method of the Lagrangean (for equality constraints) and Kuhn-Tucker conditions (for inequality constraints).

Equality Constraints: Lagrangean Method

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ for each $i = 1, \dots, k$ be $\mathcal{C}^1$ and consider the problem

$$\max f(x) \quad \text{s.t.} \quad g_i(x) = 0, \quad i = 1, \dots, k$$

- ① Form the Lagrangean (λ_i 's are called **Lagrange multipliers**)

$$L(x, \lambda) = f(x) + \sum_{i=1}^k \lambda_i g_i(x)$$

- ② Find critical points of $L(x, \lambda)$ by solving the system:

$$\frac{\partial L(x, \lambda)}{\partial x_j} = 0, \quad j = 1, \dots, n$$

$$\frac{\partial L(x, \lambda)}{\partial \lambda_i} = 0, \quad i = 1, \dots, k$$

- ③ Evaluate f at each critical point x , choose the one that gives the highest value.

Proposition A.24

Suppose that

- ① *A global optimum x^* exists.*
- ② *Constraint qualification is met at x^**

Then, there exists λ^ such that (x^*, λ^*) is a critical point of L .*

- ① This result implies that if these conditions hold, then the Lagrangean method will identify the optimum.
- ② Let $G : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be the function defined by the constraints: $G(x) \mapsto (g_1(x), \dots, g_k(x))$. The constraint qualification is the condition that the rank of the matrix $DG(x^*)$ (whose ij th element is $\partial g_i(x^*)/\partial x_j$) is equal to k . In a problem with two variables x_1, x_2 and only one constraint, this is equivalent to $\partial g_1(x^*)/\partial x_1 \neq 0$ or $\partial g_1(x^*)/\partial x_2 \neq 0$.
- ③ In many cases both conditions can be verified to hold beforehand and the method can be safely applied.

Example A.25

$$\max f(x, y) = x^2 - y^2 \text{ subject to } g(x, y) = 1 - x^2 - y^2 = 0$$

Solution

- ① Since f is continuous and the constraint set is compact there is a global maximum.
- ② $Dg(x, y) = (-2x, -2y)$ and since any solution needs to satisfy the constraint, the point $(0, 0)$ will never be an optimal solution since $0^2 + 0^2 \neq 1$, which implies that $Dg(x^*, y^*) \neq (0, 0)$ at any critical point x^* . Therefore, the constraint qualification is satisfied.

Now we can apply the Lagrangean method:

Solution (continued)

- 1 Set up the Lagrangean

$$L(x, y, \lambda) = x^2 - y^2 + \lambda(1 - x^2 - y^2)$$

- 2 Find the critical points:

$$\frac{\partial L(x, y, \lambda)}{\partial x} = 2x - 2\lambda x = 0$$

$$\frac{\partial L(x, y, \lambda)}{\partial y} = -2y - 2\lambda y = 0$$

$$\frac{\partial L(x, y, \lambda)}{\partial \lambda} = 1 - x^2 - y^2 = 0$$

Note that it must be the case that $\lambda = 1$ or $\lambda = -1$, otherwise the first two equations imply $x = y = 0$, which contradicts the third equation. Using this fact then it is easy to show that there are four possible solutions: $(x, y, \lambda) \in \{(1, 0, 1), (-1, 0, 1), (0, 1, -1), (0, -1, -1)\}$.

Solutions (continued)

- ③ Evaluate f at the critical points:

$$f(1, 0) = f(-1, 0) = 1, \quad f(0, 1) = f(0, -1) = -1$$

We conclude that the first two points are global maximizers.

Inequality Constraints: Kuhn-Tucker Method

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h_i : \mathbb{R}^n \rightarrow \mathbb{R}$ for each $i = 1, \dots, \ell$ be $\mathcal{C}^1$ and consider the problem

$$\max f(x) \text{ s.t. } h_i(x) \geq 0, \quad i = 1, \dots, \ell$$

- 1 Form the Lagrangean

$$L(x, \lambda) = f(x) + \sum_{i=1}^{\ell} \lambda_i h_i(x)$$

- 2 Find critical points of $L(x, \lambda)$

$$\frac{\partial L(x, \lambda)}{\partial x_j} = 0, \quad j = 1, \dots, n$$

$$\frac{\partial L(x, \lambda)}{\partial \lambda_i} \geq 0, \quad \lambda_i \geq 0, \quad \lambda_i \frac{\partial L(x, \lambda)}{\partial \lambda_i} = 0, \quad i = 1, \dots, \ell$$

- 3 Evaluate f at each critical point x , choose the one that gives the highest value.

Proposition A.26

Suppose that

- ① *A global optimum x^* exists.*
- ② *Constraint qualification is met at x^* .*

Then, there exists λ^ such that (x^*, λ^*) is a critical point of L .*

- Let $E \subset \{1, \dots, \ell\}$ be the subset of indexes of the constraints that hold as an equality, i.e., that are **active**. Let h_E be the E -vector of the h_i that are active. The constraint qualification is the condition that the matrix $Dh_E(x^*)$ has rank equal to E .
- Again, in many cases both conditions can be verified to hold before hand and the method can be safely applied.

Example A.27

$\max f(x, y) = x^2 - y$ subject to $g(x, y) = 1 - x^2 - y^2 \geq 0$.

Solution

- 1 Since f is continuous and the constraint set is compact there is a global maximum.
- 2 When the constraint **binds**: $x^2 + y^2 = 1$, then $x \neq 0$ or $y \neq 0$. Therefore, when the constraint is active $Dg(x^*, y^*) \neq (0, 0)$ at any critical point for which the constraint binds. Therefore, the constraint qualification is satisfied.

Now we can apply the method of Kuhn-Tucker.

Solution (continued)

- 1 Set up the Lagrangean

$$L(x, y, \lambda) = x^2 - y + \lambda(1 - x^2 - y^2)$$

- 2 Find the critical points:

$$2x - 2\lambda x = 0$$

$$-1 - 2\lambda y = 0$$

$$1 - x^2 - y^2 \geq 0, \quad \lambda \geq 0, \quad \lambda(1 - x^2 - y^2) = 0$$

From the first equation $\lambda = 1$ or $x = 0$. If $\lambda = 1$, then from the second equation $y = -1/2$, and from the constraint $x^2 + y^2 = 1$ we obtain two critical points $(\pm\sqrt{3}/2, -1/2, 1)$. If $x = 0$, then from the second equation and the slackness condition $\lambda \geq 0$ we get that $\lambda > 0$. Then we must have $x^2 + y^2 = 1$, and hence $y = \pm 1$. But $y = 1$ contradicts the second equation and $\lambda \geq 0$, and hence the only possible critical value is $(0, -1, 1/2)$.

Solution (continued)

- ③ Evaluate f at the critical points

$$f(\sqrt{3}/2, -1/2) = f(-\sqrt{3}/2, -1/2) = 2 \quad f(0, -1) = 1.$$

We conclude that the first two points are global maximizers.

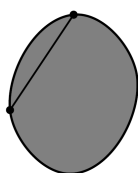
Convexity and Optimization

Under suitable convexity assumptions necessary conditions for local optima become sufficient for global optima.

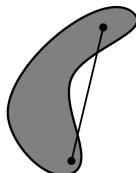
- For maximization: concave objective function and convex constraint set.
- For minimization: convex objective function and convex constraint set.

Definition A.28

A set $\mathcal{D} \subset \mathbb{R}^n$ is **convex** if for all $x, y \in \mathcal{D}$ and $\lambda \in (0, 1)$, it is the case that $\lambda x + (1 - \lambda)y \in \mathcal{D}$.



Convex



Not Convex

Let $\mathcal{D} \subset \mathbb{R}^n$ and $f : \mathcal{D} \rightarrow \mathbb{R}$. Assume from this point onward that $\mathcal{D}$ is convex.

Definition A.29

A function $f : \mathcal{D} \rightarrow \mathbb{R}$ is **concave** if for any $x, y \in \mathcal{D}$ and $\lambda \in (0, 1)$, it is the case that

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$$

and **convex** if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Definition A.30

A function $f : \mathcal{D} \rightarrow \mathbb{R}$ is **strictly concave** if for any $x, y \in \mathcal{D}$ such that $x \neq y$ and $\lambda \in (0, 1)$, it is the case that

$$f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y)$$

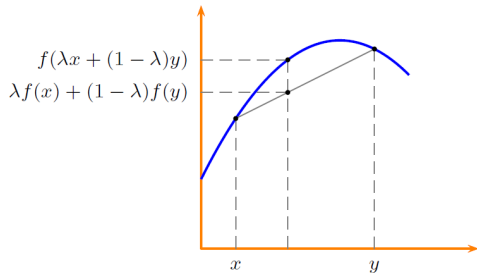
and **strictly convex** if

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y)$$

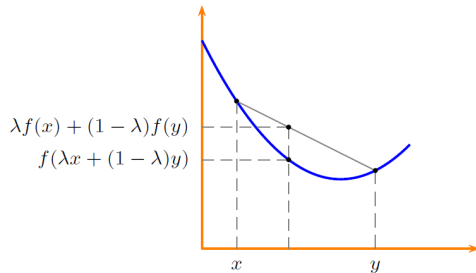
Theorem A.31

$f : \mathcal{D} \rightarrow \mathbb{R}$ is (strictly) concave if and only if $-f$ is (strictly) (convex).

Concave and Convex Functions



A concave function



A convex function

Definition A.32

A **quadratic form** in n variables is a function $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ for which

$$Q(x_1, \dots, x_n) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j = a_{11} x_1^2 + a_{12} x_1 x_2 + \cdots a_{ij} x_i x_j + \cdots + a_{nn} x_n^2$$

for all $(x_1, \dots, x_n)$, where a_{ij} for $i = 1, \dots, n$, and $j = 1, \dots, n$ are constants.

Note that we can arrange the constants a_{ij} in a $n \times n$ *symmetric* matrix $\mathbf{B}$ where the index i represents the rows and the index j represents the columns, where the (i, j) element is $b_{ij} = \frac{1}{2}(a_{ij} + a_{ji})$. Then, we can write any quadratic form as

$$Q(x_1, \dots, x_n) = \mathbf{x}^\top \mathbf{B} \mathbf{x}$$

where $\mathbf{x}$ is the column vector with components $(x_1, \dots, x_n)$ and $\mathbf{x}^\top$ is its transpose, i.e., a row vector.

Definition A.33

Let Q be a quadratic form and let $\mathbf{B}$ be the symmetric matrix that represents it. Then Q (and the associated matrix $\mathbf{B}$) is

- **Positive definite** if $x^\top \mathbf{B}x > 0$ for all $x \neq 0$.
- **Negative definite** if $x^\top \mathbf{B}x < 0$ for all $x \neq 0$.
- **Positive semidefinite** if $x^\top \mathbf{B}x \geq 0$ for all x .
- **Negative semidefinite** if $x^\top \mathbf{B}x \leq 0$ for all x .
- **Indefinite** if it is neither positive nor negative semidefinite.

Remark: When talking about whether a quadratic form satisfies any of the above definitions, we usually use the above concepts to talk about the associated matrix $\mathbf{B}$.

There is a convenient procedure to check whether $\mathbf{B}$ satisfies any of the above, check the following links:

- [Quadratic forms: conditions for definiteness](#)
- [Quadratic forms: conditions for semidefiniteness](#)

Concave and Convex Functions

Note that the Hessian matrix $D^2f(x)$ can be used to define a quadratic form. Then we can characterize concavity or convexity by looking at $D^2f(x)$.

Theorem A.34

Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be a $\mathcal{C}^2$ function, where $\mathcal{D} \subset \mathbb{R}^n$ is open and convex. Then,

- ① f is concave if and only if $D^2f(x)$ is negative semidefinite for all $x \in \mathcal{D}$.
- ② f is convex if and only if $D^2f(x)$ is positive semidefinite for all $x \in \mathcal{D}$.
- ③ If $D^2f(x)$ is negative definite for all $x \in \mathcal{D}$, then f is strictly concave.
- ④ If $D^2f(x)$ is positive definite for all $x \in \mathcal{D}$, then f is strictly convex.

Be careful!

Note that points 3 and 4 above **do not** claim that if f is strictly concave (convex), then $D^2f(x)$ is negative definite (positive definite) for all $x \in \mathcal{D}$. Consider for example the function $f(x) = -x^4$. Note that f is strictly concave but $D^2f(0) = f''(0) = 0$, so $D^2f(0)$ is not negative definite.

Procedure for checking the concavity/convexity and strict concavity/convexity of a function of many variables

- 1 Hessian negative definite for all $x \Rightarrow$ function is strictly concave; Hessian positive definite for all $x \Rightarrow$ function is strictly convex.
- 2 Hessian not negative semidefinite for all $x \Rightarrow$ function is not concave, and hence not strictly concave; Hessian not positive semidefinite for all $x \Rightarrow$ function is not convex, and hence not strictly convex.
- 3 Hessian negative semidefinite for all x but not negative definite for all $x \Rightarrow$ function is concave and may or may not be strictly concave; you need to use other methods (for example, the basic definition of concavity) to determine whether it is strictly concave. Similarly, Hessian positive semidefinite for all x but not positive definite for all $x \Rightarrow$ function is convex and may or may not be strictly convex.

Concave and Convex functions

Recall that $f_{ij}(x) = \partial^2 f(x) / \partial x_j \partial x_i$.

- A function of one variable $f(x)$ is concave iff $f''(x) \leq 0$.
- A function of two variables $f(x, y)$ is concave iff

$$f_{11} \leq 0, \quad f_{22} \leq 0, \quad f_{11}f_{22} - f_{12}^2 \geq 0$$

- If $f_{11} < 0$ and $f_{11}f_{22} - f_{12}^2$, then it is strictly concave.

Example A.35

Let $f(x) = -(x - 1)^2 + 2$. Then, $f''(x) = -2 < 0$ and hence this function is strictly concave.

Example A.36

Let $f(x) = 1 - x^2 - y^2$. Then, $f_{11} = -2 < 0$ and $f_{11}f_{22} - f_{12}^2 = (-2)(-2) - 0 > 0$, and hence this function is strictly concave.

Theorem A.37

Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be concave and $\mathcal{D}$ convex. Then,

- ① Any local maximum of f is a global maximum.
- ② The set of maximizers is either empty or convex. If f is strictly concave, then the set of maximizers is either empty or contains a single point.

Theorem A.38

Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be differentiable and concave and $\mathcal{D}$ convex. Then, x is an interior maximum of f on $\mathcal{D}$ if and only if $Df(x) = 0$.

- Let $U \subset \mathbb{R}^n$ be open and convex.
- $f : U \rightarrow \mathbb{R}$ and $h_i : U \rightarrow \mathbb{R}$ for each $i = 1, \dots, \ell$ concave and $\mathcal{C}^1$.
- Consider the problem

$$\max f(x) \quad \text{s.t.} \quad h_i(x) \geq 0, \quad i = 1, \dots, \ell$$

- If there is some $\bar{x} \in U$ such that $h_i(\bar{x}) > 0$ for all $i = 1, \dots, \ell$ (this condition is known as **Slater's condition**, then the Kuhn-Tucker method will always identify global optima.
- No need to verify
 - existence of a solution
 - constraint qualification

- ① If L has no critical point, then there is no solution to the problem.
- ② If (x^*, λ^*) is a critical point, then it is a solution.

Also

- If Slater's condition is not met, then the second point is still true, whereas the first is not.
- If f is strictly concave, then there exists at most one solution.

Example A.39

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and consider the problem:

$$\max f(x) = -(x - 2) \text{ subject to } h(x) = x - 1 \geq 0.$$

Solution

- ❶ $f'(x) = -2(x - 2)$ and $f''(x) = -2 < 0$, and hence f is (strictly) concave; $h'(x) = 0$ and hence h is concave.
- ❷ There is an $x \in \mathbb{R}$ (e.g., $x = 2$) such that $h(x) > 0$.

This implies

- ❶ If the Lagrangean has no critical point, then there is no solution to the problem.
- ❷ If (x^*, λ^*) is a critical point, then it is a solution. In fact, since f is strictly concave, it is the unique solution.

Solution (continued)

- ① Set up the Lagrangean:

$$L(x, \lambda) = -(x - 2)^2 + \lambda(x - 1)$$

- ② Find the critical points:

$$-2(x - 2) + \lambda = 0$$

$$x - 1 \geq 0, \quad \lambda \geq 0, \quad \lambda(x - 1) = 0.$$

- First equation and $\lambda \geq 0 \Rightarrow x \geq 2$.
- $\lambda(x - 1) = 0 \Rightarrow \lambda = 0$.
- First equation $\Rightarrow x = 2$.

Therefore, the only critical point is $(2, 0)$, and $x = 2$ is the unique global maximizer.

Derivatives of functions defined implicitly.

- Suppose that an economic model delivers the following *equilibrium equation*

$$f(x, p) = 0$$

where x is an economic variable of interest and p is a parameter.

- Important question: How does the equilibrium value of x varies with p ?
 - Can we answer this question with only minimal assumptions on f ?
 - Typically, we cannot or do not want to solve *explicitly* for x as a function of p .
- We must first ask the question if for each value of p the equation has a solution.
 - One tool: use the **intermediate value theorem**: if f is continuous and we have that for all $x \in (x_1, x_2)$ and we have that $f(x_1, p) < 0$ and $f(x_2, p) > 0$ (or alternatively $f(x_1, p) > 0$ and $f(x_2, p) < 0$), then the IVT tells us that there exists $x^* \in (x_1, x_2)$ for which $f(x^*, p) = 0$.
 - Even if the f does not satisfy the conditions of the IVT, we may be able to argue that a solution exists by using the specific properties of f .

Derivatives of functions defined implicitly.

- How does the solution, if it exists, depends on p ?
- If x_0 is a solution for the parameter p_0 , so that $f(x_0, p_0) = 0$, and the function g that the equation $f(g(p), p) = 0$ **implicitly defines** is differentiable at p_0 , then we can find its derivative $g'(p)$ by differentiating both sides of the equation:

$$f_1(g(p), p)g'(p) + f_2(g(p), p) = 0 \Rightarrow g'(p) = -\frac{f_2(g(p), p)}{f_1(g(p), p)}$$

whenever $f_1(g(p), p) \neq 0$.

Theorem A.40 (Implicit Function Theorem (for 2 variables))

Let f be a continuously differentiable function of two variables defined on an open set S . If $f_1(x_0, p_0) \neq 0$, then there exists a continuously differentiable function g of a single variable defined on an open interval I containing p_0 such that $f(g(p), p) = f(x_0, p_0)$ for all $p \in I$, and

$$g'(p_0) = -\frac{f_2(x_0, p_0)}{f_1(x_0, p_0)}.$$

Application: slopes of indifference curves

Let $U(x, y)$ be a utility function. The equation $U(x, y) = c$ where c is a constant defines an **indifference curve**.

- 1 The equation $U(x, y) = c$ defines, for each c , y as a function of x , i.e., $U(x, g(x)) = c$ for all x that satisfy the equation.
- 2 By the Implicit Function Theorem, if $U(x_0, y_0) = c$, the slope of the level curve through (x_0, y_0) at this point is given by

$$g'(x_0) = -\frac{U_1(x_0, y_0)}{U_2(x_0, y_0)}$$

- 3 Note that we are expressing the second argument as a function of the first argument, rather than the first as a function of the second, as stated in the theorem. Thus, the indices 1 and 2 have to be interchanged.
- 4 Note that $g'(x_0)$ is the well-known **marginal rate of substitution at (x_0, y_0)** .

The General Implicit Function Theorem.

- Suppose now that we have an economic model with n endogenous variables $\mathbf{x} = (x_1, \dots, x_n)$ and m exogenous parameters $\mathbf{p} = (p_1, \dots, p_m)$. Suppose the equilibrium of such a model can be described by the system of equations:

$$f^1(\mathbf{x}, \mathbf{p}) = 0$$

$$\vdots$$

$$f^n(\mathbf{x}, \mathbf{p}) = 0$$

with $f^i : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ for $i = 1, \dots, n$. Note that there is one equation for each unknown.

- Note that if we define the vector-valued function $\mathbf{F} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, where $\mathbf{F}(\mathbf{x}, \mathbf{p}) = (f^i(\mathbf{x}, \mathbf{p}))_{i=1, \dots, n}$, the above system can be written compactly as

$$\mathbf{F}(\mathbf{x}, \mathbf{p}) = \mathbf{0},$$

where $\mathbf{0}$ is the zero-vector in $\mathbb{R}^n$.

The General Implicit Function Theorem.

- How does the solution, if it exists, depends on $\mathbf{p}$?
- A solution, if it exists can be expressed as a vector-valued function $\mathbf{g} : \mathbb{R}^m \rightarrow \mathbb{R}^n$ that determines $\mathbf{x}$ in terms of $\mathbf{p}$.
- Once again, we can compute $D\mathbf{g}$ without calculating $\mathbf{g}$ explicitly. This result is the **General Implicit Function Theorem**.
- Recall that the derivative of $\mathbf{F}$, also called the Jacobian, is given by

$$D\mathbf{F}(\mathbf{x}, \mathbf{p}) = \left[\begin{array}{ccc|ccc} \frac{\partial f^1}{\partial x_1}(\mathbf{x}, \mathbf{p}) & \cdots & \frac{\partial f^1}{\partial x_n}(\mathbf{x}, \mathbf{p}) & \frac{\partial f^1}{\partial p_1}(\mathbf{x}, \mathbf{p}) & \cdots & \frac{\partial f^1}{\partial p_m}(\mathbf{x}, \mathbf{p}) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^n}{\partial x_1}(\mathbf{x}, \mathbf{p}) & \cdots & \frac{\partial f^n}{\partial x_n}(\mathbf{x}, \mathbf{p}) & \frac{\partial f^n}{\partial p_1}(\mathbf{x}, \mathbf{p}) & \cdots & \frac{\partial f^n}{\partial p_m}(\mathbf{x}, \mathbf{p}) \end{array} \right] = [D_{\mathbf{x}}\mathbf{F}(\mathbf{x}, \mathbf{p}) | D_{\mathbf{p}}\mathbf{F}(\mathbf{x}, \mathbf{p})]$$

- The submatrices $D_{\mathbf{x}}\mathbf{F}(\mathbf{x}, \mathbf{p})$ and $D_{\mathbf{p}}\mathbf{F}(\mathbf{x}, \mathbf{p})$ are the Jacobians of $\mathbf{F}$ with respect only to the variables $\mathbf{x}$ and $\mathbf{p}$ respectively.

The General Implicit Function Theorem.

Theorem A.41

General Implicit Function Theorem. Let $\mathbf{F} : X \times P \rightarrow Y$, where $X, Y \subset \mathbb{R}^n$ and $P \subset \mathbb{R}^m$ be a $\mathcal{C}^n$ function on an open neighborhood of $(\mathbf{x}_0, \mathbf{p}_0)$ at which $D_{\mathbf{x}}\mathbf{F}(\mathbf{x}_0, \mathbf{p}_0)$ is non-singular (i.e., the matrix inverse exists) and $\mathbf{F}(\mathbf{x}_0, \mathbf{p}_0) = \mathbf{0}$. Then there exists a neighborhood $U \subset \mathbb{R}^n$ of $\mathbf{p}_0$ and a unique function $\mathbf{g} : U \rightarrow X$ such that

$$\mathbf{x}_0 = \mathbf{g}(\mathbf{p}_0) \quad \text{and} \quad \mathbf{F}(\mathbf{g}(\mathbf{p}), \mathbf{p}) = \mathbf{0} \quad \text{for every } \mathbf{p} \in U$$

Furthermore, $\mathbf{g}$ is $\mathcal{C}^n$ on U with

$$D_{\mathbf{p}}\mathbf{g}(\mathbf{p}_0) = - (D_{\mathbf{x}}\mathbf{F}(\mathbf{x}_0, \mathbf{p}_0))^{-1} D_{\mathbf{p}}\mathbf{F}(\mathbf{x}_0, \mathbf{p}_0)$$